

Handling NULL Values in SQL

COALESCE() vs ISNULL() vs CASE
WHEN

A quick reference for Data Analytics

Sample Data: Employees Table

- Emp_ID | Salary | Bonus
1 | 50000 | NULL
2 | NULL | 8000
3 | NULL | NULL
- Goal: Derive Final Pay without NULL values

COALESCE()

- Returns the first non-NULL value from a list
- Works across most databases
- Example:
COALESCE(Salary, Bonus, 0) AS Final_Pay
- Best when multiple columns may contain NULL

ISNULL()

- SQL Server specific function
- Replaces NULL with a single value
- Example:
ISNULL(Salary, 0) AS Salary
- Best for simple NULL replacement

CASE WHEN

- Used for custom business logic
- More flexible but verbose
- Example:

```
CASE WHEN Salary IS NOT NULL THEN Salary  
      WHEN Bonus IS NOT NULL THEN Bonus ELSE 0  
END
```
- Best when conditions are complex

Key Takeaways

- COALESCE() → Clean and flexible
- ISNULL() → Simple replacement (SQL Server)
- CASE WHEN → Best for business logic
- Choose based on use case, not habit